



# Network Citation

## *Visualising Bibliographic Citation Networks in Edubba Studies*

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**2,103**

Publications

**2,524**

Citation Edges

**5**

Hypotheses

# Presentation Structure

**01**

## **Introduction & Problem**

Edudda schools · Why network analysis?

**02**

## **Data & Methods**

2,103 sources · Step-Distance Model · Pipeline

**03**

## **RQ & Hypotheses**

2 research questions · 5 hypotheses

**04**

## **Results H1–H5**

5 visualizations · Design choices

**05**

## **Conclusions & Limitations**

Findings · Future work

# 01

## Introduction & Problem

*Edubba schools · The knowledge gap · Why visualisation?*

# What is Edubba — and What is the Problem?

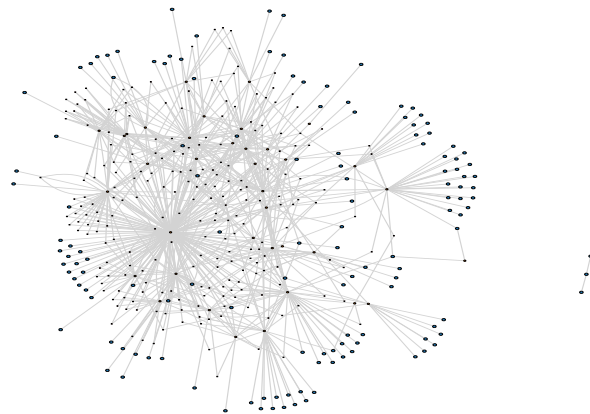
**Edubba** — one of the earliest scribal schools, 2nd millennium BCE. Evidence of structured education.

**The Problem:** knowledge is fragmented. Lack of primary and reliable sources. Many interpretations and speculations.

**The Target:** analyse how scholarly knowledge about Edubba is structurally organised and transmitted within academic citation networks.

**Approach:** 2,103 publications modelled as a directed citation network. Visual analytics reveals hidden epistemic structures.

This project approaches the research questions through information visualisation: citation relationships are encoded as a directed network, and visual analytics — node layout, colour, and centrality — reveal epistemic structures not accessible through linear reading.



# 02

## Data & Methods

*Collection · Parsing · Snowball · Graph Construction · Step-Distance Model*

# Data Collection Pipeline

1

## Collect Anchor Sources

~150 academic PDFs  
(DOI verified)

2

## Collect Primary Sources

Archaeological site data  
(CSV, open source)

3

## Parse & Clean

pdfplumber · OCR  
Metadata → CSV

4

## Snowball Expansion

AnyStyle bibliography  
extraction → JSON

5

## Source Merging

2,103 unified sources  
138 anchor nodes

6

## Graph Construction

igraph · 2,103 nodes  
2,524 edges

# Step-Distance Model — Epistemic Classification

*Each source assigned an epistemic distance from primary data (Step 1–5)*

| Step   | Role            | Criteria                                                                  | N     | %     |
|--------|-----------------|---------------------------------------------------------------------------|-------|-------|
| Step 1 | Direct Reliance | Archaeological pubs — direct empirical evidence (manual)                  | 164   | 7.8%  |
| Step 2 | Mediators       | betweenness $\geq$ top 15% AND cites $\geq$ 1 Step 1 node                 | 37    | 1.8%  |
| Step 3 | Authorities     | in-degree $\geq$ top 10% AND cited by $\geq$ 2 Step2 AND age $\geq$ 10 yr | 226   | 10.7% |
| Step 4 | Mediated        | cites no Step1 AND cites $\geq$ 2 Step3 AND out-degree $\geq$ 2           | 3     | 0.1%  |
| Step 5 | Peripheral      | All others — NOT Step 1/2/3/4                                             | 1,673 | 79.6% |

# 03

## Research Questions & Hypotheses

*2 RQs · 5 Hypotheses forming a chain of evidence*



# Research Questions & Hypotheses Chain

**RQ1**

How is scholarly knowledge about Edubba schools structurally organised within the citation network — and what role do primary archaeological sources play?

**RQ2**

Which sources can be identified as epistemically reliable based on their structural position relative to primary data — and do the most reliable sources coincide with the most cited?

*Operationalised through 5 sequential hypotheses:*

**H1**

**Network Topology**

*Hierarchical structure?*

**H2**

**Structural Gap**

*Step1 vs Step3 isolated?*

**H3**

**Bottleneck**

*Step 2 — only bridge?*

**H4**

**Epistemic Displacement**

*Eigenvector in Step 3?*

**H5**

**Reliability Rarity**

*Trust Triangle < 1%?*

# 04

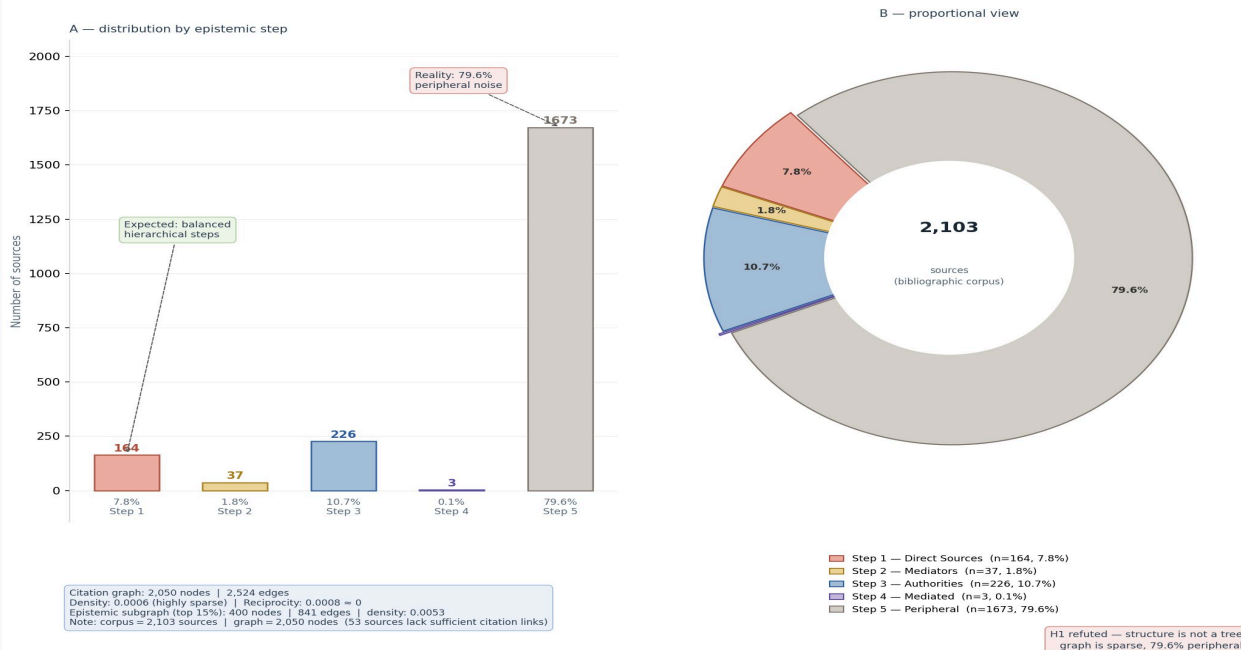
## Results H1–H5

*5 visualizations · Design rationale · Key findings*

Is knowledge about Edubba distributed as a hierarchy — roots (Step 1) → branches (Step 2/3)?

## H1 — Is Knowledge Distribution Hierarchical (Tree-like)?

Bibliographic corpus: 2,103 sources | Citation graph: 2,050 nodes



Sources: sources\_withAll\_steps\_v2\_LABELLED.csv | All publications uniquely identifiable via DOI.

## How to Read

Read Panel A left to right: Step 1 through Step 5. A hierarchical structure would produce a roughly bell-shaped or pyramid distribution — comparable bars for Steps 1–4 with Step 5 smaller. Instead, the bar for Step 5 dominates visually (n = 1,673, 79.6%), while Steps 2 and 4 are barely visible. Panel B reinforces this: the near-total grey segment of Step 5 makes the imbalance immediately apparent. The green annotation marks what would be expected in a hierarchy, the red annotation marks the actual result

**79.6%**  
Step 5 — Peripheral

**0.0006**  
Network Density

**1.8%**  
Step 2 — Mediators only

Design: Bar chart (position on common scale) for Compare task · Donut for part-of-whole · (Cleveland & McGill, 1984)

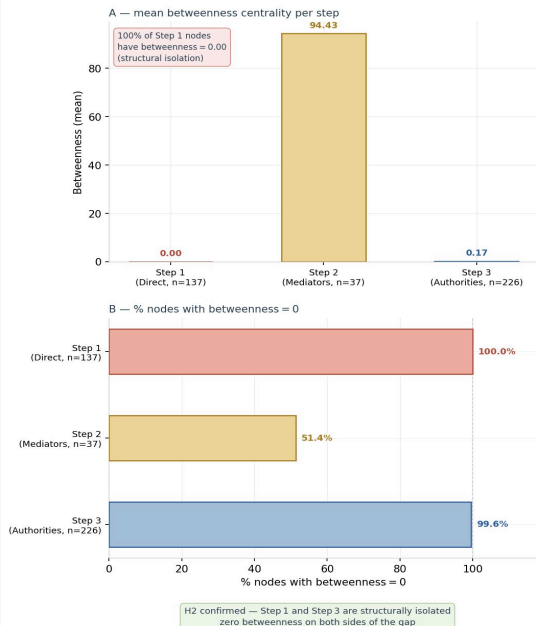
**Key Finding: network is NOT a tree — sparse, scattered, 79.6% peripheral**

Source: sources\_withAll\_steps\_v2\_LABELLED.csv | citation\_graph.csv

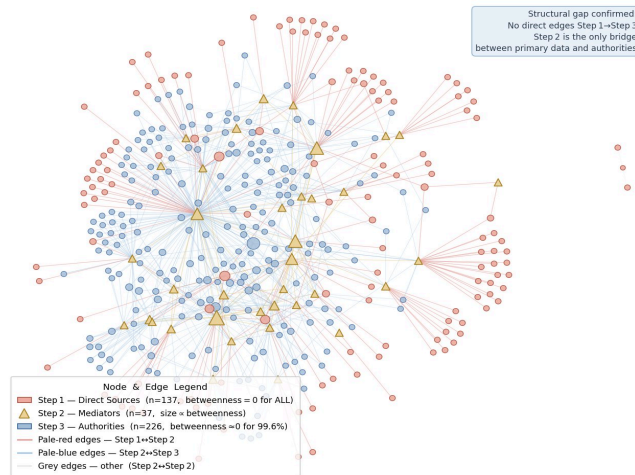
Are primary archaeological sources (Step 1) structurally isolated from authoritative works (Step 3)?

### H2 — Are Step 1 (primary) and Step 3 (authorities) structurally disconnected?

Epistemic subgraph, N = 400 nodes



**C — epistemic citation network coloured by step**  
(no direct edges Step 1 → Step 3 — Step 2 is the only bridge)



Source: epistemic\_subgraph\_with\_centralities2.pdf | All publications uniquely identifiable via DOI.

## How to Read

In Panel A: compare bar heights across the three steps. Step 2 (pale amber bar) should be visibly taller.

In Panel B: the near-total fill for Steps 1 and 3 (100% and 99.6% betweenness = 0) versus Step 2 confirms structural isolation.

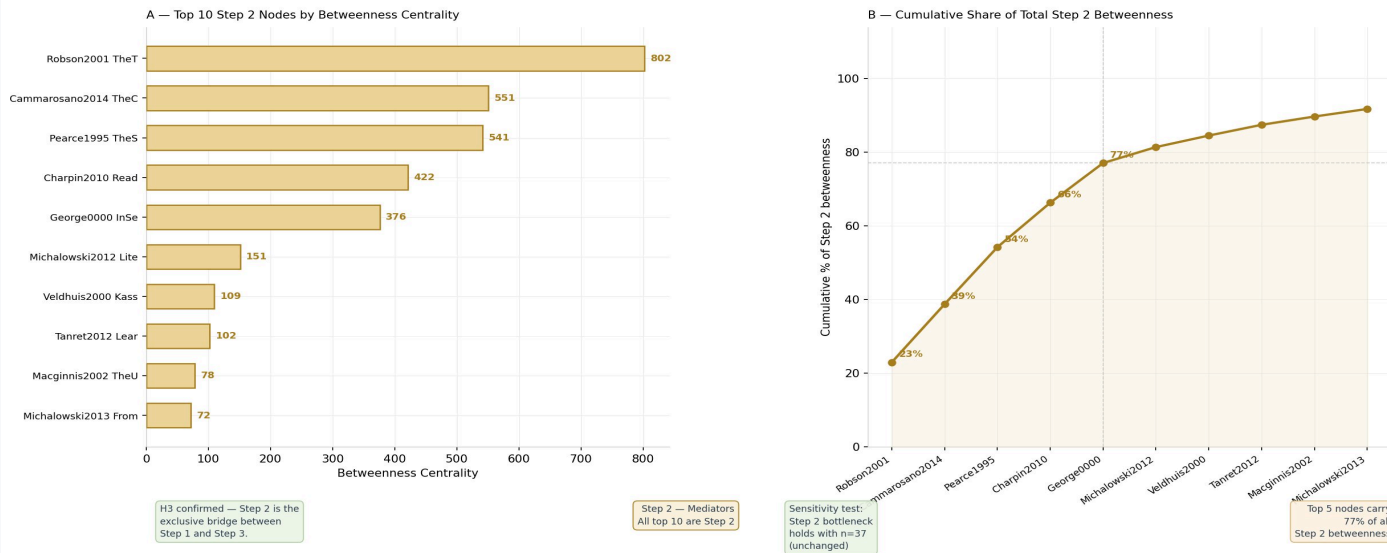
In Panel C: locate the pale terracotta (red) cluster of Step 1 nodes and the pale blue cluster of Step 3 nodes — observe the spatial separation. Pale amber triangles (Step 2) act as the only visible bridge..

**All 137 Step 1 nodes:** betweenness = 0.00.  
Primary sources are cited but never transit information.  
No direct connections  
Step 1 → Step 3 exist

Is Step 2 the only structural bridge between Step 1 and Step 3? Is it robust to threshold changes?

### H<sub>3</sub> — Does Step 2 Form the Exclusive Bottleneck Between Step 1 and Step 3?

Epistemic subgraph, N = 400 | Step 2: n=37 (1.8% of full corpus)



## How to Read

Panel A: read bars left to right — single amber colour confirms all top-10 betweenness nodes are Step 2.  
Panel B: the steeper the cumulative curve, the more concentrated the bottleneck. Top 5 mediators carry 77% of all Step 2 betweenness.

**37 nodes (1.8%)** = entire bridge Step 1 → Step 3. Top 5 carry 77% of all Step 2 betweenness.

**Sensitivity test:** With filter = 37 · Without filter = 37. Bottleneck is a REAL structural property, not a threshold artifact.

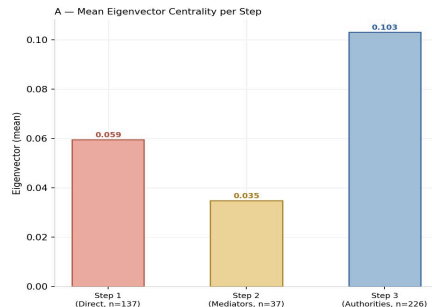
Sources: epistemic\_subgraph\_with\_centralities2.pkl | All publications uniquely identifiable via DOI.

Design: Horizontal bar (Panel A — Identify Extremes, long source IDs) · Cumulative line (Panel B — concentration, Lorenz curve analogy)

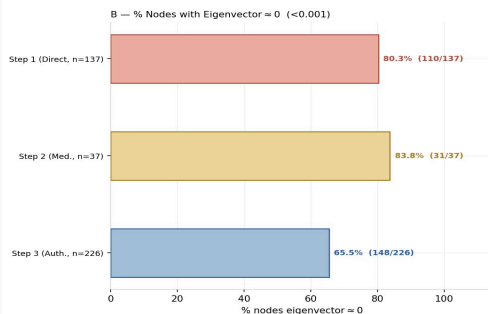
*Is eigenvector centrality concentrated in Step 3 rather than Step 1? Does influence decouple from proximity to primary data?*

#### H4 — Where Does Structural Influence (Eigenvector) Concentrate?

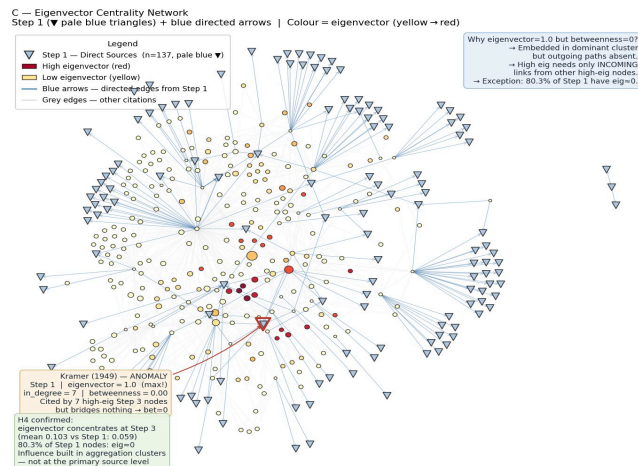
Epistemic subgraph, N = 400 | Eigenvector centrality by step



Step 3 mean is 1.73x higher than Step 1 | Step 2 is LOWER than Step 1



73%+ of all nodes across ALL steps have eigenvector = 0



Sources: epistemic\_subgraph\_with\_centralities2.pdf | All publications uniquely identifiable via DOI.

## How to Read

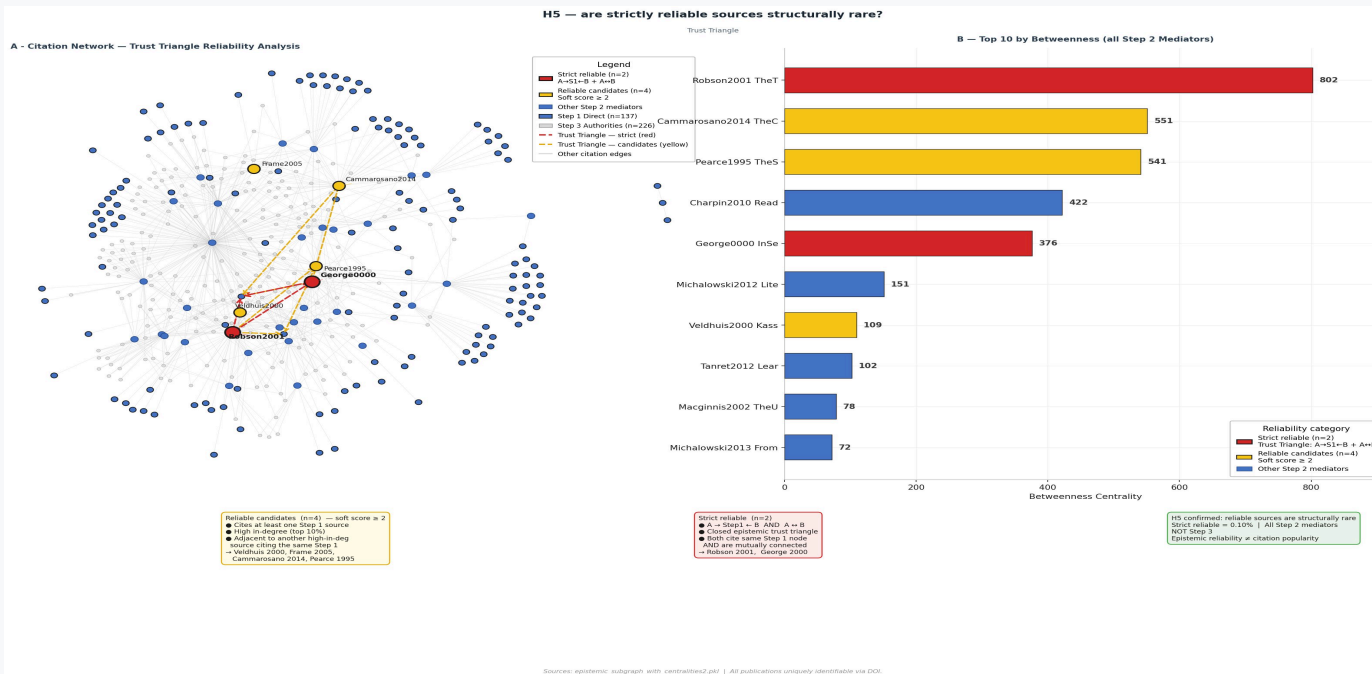
In Panel C: Network Heatmap warm-colored (yellow/red) nodes represent high-eigenvector publications, which demonstrate that most popular sources (red nodes) are not connected to primary sources (blue triangle). The blue triangles consistently appear in cool colors (yellow/pale) at the periphery. The one exception is Kramer1949, annotated explicitly.

**Anomaly — Kramer (1949):** Step 1 node with eigenvector = 1.0 (max!), cited by 7 central Step 3 nodes, but betweenness = 0. A dead end, not a bridge.

**Step 2 Paradox:** Mediators have the LOWEST mean eigenvector (0.035) — even below Step 1 (0.059). Bridges connect isolated zones without belonging to the epistemic core.

Design: Network heatmap (Panel C) encodes topology + scalar intensity (YlOrRd). Scatter plot rejected — loses spatial/topological dimension.

Are strictly reliable sources (primary link + high in-degree + mutual reinforcement) < 1% of corpus, all Step 2?



## How to Read

-Two red nodes (Robson2001 and George0000) = strictly reliable sources. Three dashed red edges form a visible triangle via a shared Step 1 node.

-Yellow nodes = top-4 soft candidates — no closed triangle.

Vast majority of nodes are grey or blue, confirming reliability is structurally rare.

## Strictly reliable:

Link to primary data, High in-degree. There is another node B with a high in-degree that cites the same Step 1 source as A, forming a closed trust triangle.

## Key conclusion:

< 0.1%

all are Step 2

## Share of corpus:

Reliability != Popularity  
Adams1969 (in-deg=21) FAILED

Design: Node-link required — Trust Triangle is a topological property (closed triad). Dashed edges pre-attentively group the triangle. Bar chart (Panel B) confirms all are Step 2.

# Unified Design System — Visual Grammar

*Consistent color encoding + layout across all 5 figures enables cross-figure visual comparison*

## Color Encoding (Step-Distance)



### Step 1 — Direct Reliance (Primary)

Pale Terracotta (Melon) · H=10° S=68% L=78% · #EDADA0



### Step 2 — Mediators (Bottleneck)

Pale Amber (Jasmine) · H=42° S=70% L=76% · #EDD498



### Step 3 — Authorities (High in-degree)

Pale Blue (Beau Blue) · H=210° S=40% L=75% · #A4BED8



### Step 4 — Mediated

Pale Lavender · H=260° S=34% L=79% · #C4B8DC



### Step 5 — Peripheral (Noise)

Pale Silver · H=36° S=10% L=80% · #D2CEC8

## Design Principles Applied

- Position on scale for Compare (Cleveland & McGill, 1984)
- Horizontal bars for long source identifiers
- Node-link graphs for topological properties
- Sequential color scale (YlOrRd) for eigenvector heatmap
- White background, removed top/right spines (Tufte)
- Semi-transparent annotation boxes (alpha = 0.8)
- DejaVu Sans — consistent cross-platform font

### Bar Chart

H1, H2, H3, H4  
Compare task

### Donut Chart

H1  
Part-of-whole

### Node-Link Graph

H2, H3, H5  
Topology

### Network Heatmap

H4  
Topology + scalar

### Cumulative Line

H3  
Concentration



# 05

## Conclusions & Limitations

*Findings · Methodological contributions · Future work*

# Key Findings & Conclusions

1

## H1 — Not hierarchical

79.6% of sources are peripheral. Citation network is sparse and fragmented — not tree-like. Visual analytics immediately contradicts the hierarchical model.

2

## H2 — Primary sources are isolated

Step 1 betweenness = 0.00. Primary data is cited but never mediates information flow. The gap between evidence and interpretation is structural.

3

## H3 — Bottleneck confirmed

37 nodes (1.8%) carry all epistemic traffic Step 1 → Step 3. Robust to threshold changes. Top 5 mediators carry 77% of all betweenness.

4

## H4 — Influence is not proximity

Eigenvector concentrates in Step 3, not Step 1. Discursive authority is decoupled from empirical proximity. Step 2 paradox: bridges without belonging.

5

## H5 — Reliability is rare and unexpected

Only 2 sources satisfy all Trust Triangle criteria. Both are Step 2 Mediators — NOT the most cited Step 3 Authorities. Citation popularity and epistemic reliability are structurally different properties.

**Core contribution:** Visual analytics reveals hidden epistemic structures in humanities research — structures not accessible through linear reading or citation counts alone.

# Limitations & Future Work

## Limitations

- Node-link diagrams do not scale to the full 2,103-node graph — subgraph (N=400) used for topology
- Step-Distance Model assigns roles by network position, not content validity
- Step 4 is under-represented in the dataset, with only three identified sources. This is likely due to the DOI-based sampling strategy, which favours journal articles over textbooks and encyclopedic works. As a result, the visualisations may under-represent synthesis-level scholarship. The Step 4 layer in the figures should therefore be interpreted as a partial reflection of the epistemic landscape rather than a complete one
- The node-link diagram is useful for small subgraphs but does not scale to the full dataset, because visual complexity increases faster than human perceptual capacity. Therefore, alternative representations are required for global structure analysis
- Network reflects the selected corpus only — not a comprehensive field map

## Future Work

- Incorporate temporal dynamics of citation formation
- Test Trust Triangle robustness under alternative reliability definitions
- Compare Edubba network with other archaeological/historical domains
- Expand corpus to include grey literature and monographs
- Content analysis alongside structural analysis

# Thank You

## *Network Citation: Visualising Bibliographic Citation Networks in Edubba Studies*

### Results Summary:

H1 NOT CONFIRMED — Not hierarchical (79.6% peripheral)

H2 CONFIRMED — Primary sources structurally isolated

H3 CONFIRMED — 37 nodes (1.8%) = entire bottleneck

H4 CONFIRMED\* — Eigenvector in Step 3, not Step 1

H5 CONFIRMED — Only 2 reliable sources, both Step 2

### Data & Code

[github.com/AimBtlv/  
research-NetworkCitation](https://github.com/AimBtlv/research-NetworkCitation) Edduba

Folder: 8-output\_Modul2

Tools: Python · igraph · AnyStyle  
pdfplumber · Tableau